

Exact localization of Grover quantum walks on crystal lattices: transfer currents, a universal bound, and uniform-spanning-tree degree laws

Jacob Goodchild*

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Discrete-time Grover walks on periodic graphs localize: a finite fraction of the wave function stays at the starting vertex forever because the walk operator has flat (momentum-independent) eigenvalues. We compute the long-time average return probability $\bar{p}$ exactly for a large family of lattices. For the moving-shift walk on $\mathbb{Z}^d$ we show that $\bar{p}$ is governed by $A_d = \int_0^\infty \operatorname{erfc}(\sqrt{s})^d ds$, and we obtain $A_3 = \frac{1}{2} - (3 - \sqrt{3})/\pi$, $A_4 = \frac{1}{2} - (18 - 8\sqrt{3})/(3\pi)$ and an arccosine closed form for A_5 . For the flip-flop walk we prove that the ± 1 flat-band projectors are $I - Y$ and $I - Y^Q$, where Y and Y^Q are the Kirchhoff transfer-current matrices of the Laplacian and signless Laplacian. This reduces the trapping probability to lattice Green functions. It gives closed forms in terms of Watson integrals (simple cubic, body-centred cubic, face-centred cubic), complete elliptic integrals (triangular, kagome, checkerboard, star, line graphs), and elementary functions of the bond anisotropy. We prove the universal lower bound $\bar{p} \geq (d-2)^2/[2d(d-1)]$ for edge-transitive d -regular lattices, where d is the vertex degree. The bound is attained exactly on bipartite lattices whose vertex stabilizer acts 2-transitively on the incident edges (honeycomb, diamond, the Laves graph, d -dimensional diamond). We also solve lazy (Szegedy) versions on the square, triangular, honeycomb and diamond lattices as explicit functions of the laziness. The same transfer-current matrix yields the vertex-degree distribution of the uniform spanning tree. On 2-arc-transitive lattices it is exactly $1 + \operatorname{Binomial}(z-1, 1/(z-1))$, and we give closed forms for the simple-cubic, BCC, FCC, pyrochlore, king and snub-square lattices. All closed forms were identified by integer-relation (PSLQ) searches at 25–80 digits and checked independently by Brillouin-zone diagonalization, direct simulation, or both.

I. INTRODUCTION

Quantum walks are the unitary analogues of classical random walks [1]. They underpin quantum search algorithms, in particular Szegedy’s quantization of Markov chains [2] and the lackadaisical (lazy) walks introduced by Wong [3]. A striking feature of the Grover walk in two or more dimensions is *localization*: even on a perfect lattice, the probability of finding the walker at its starting site does not decay to zero. Inui, Konishi and Konno established this for the two-dimensional Grover walk and computed the limiting return probability [4]. Higuchi, Konno, Sato and Segawa related localization of Grover walks on crystal lattices to flat bands of the walk operator and gave integral descriptions in any dimension [5]. What has been missing is a systematic set of *closed forms* beyond the square lattice.

This paper supplies them. The main structural observation (Sec. II) is that for the flip-flop Grover walk the $+1$ and -1 flat-band projectors are the projectors onto the cycle spaces of the Laplacian $L = D - A$ and of the signless Laplacian $Q = D + A$. Their matrix elements between arcs at a vertex are therefore Kirchhoff transfer currents [6, 7], the same objects that give edge correlations of the uniform spanning tree (UST). Three consequences follow. (i) Exact trapping probabilities reduce to lattice Green functions at the two band edges. These are classical in several cases: Watson integrals [8, 9], Joyce’s cubic Green functions [10, 11] and

Horiguchi’s triangular and honeycomb formulas [12]. (ii) Foster’s theorem [13] and the Cauchy–Schwarz inequality give a universal lower bound, attained on 2-arc-transitive bipartite lattices (Sec. III). (iii) The UST degree distribution at a vertex is $\det[I + (x-1)Y]$ [7], which gives an exact binomial law in the symmetric case (Sec. V).

We also treat the moving-shift walk on $\mathbb{Z}^d$ in any dimension (Sec. IV). There the trapping integral is a Cauchy or Gaussian average that closes in elementary terms for $d \leq 5$.

Status of novelty. Every result below was checked against the literature by web search. Items we believe are already known, or probably known, are flagged explicitly: the two-dimensional moving-shift values [4], the integral A_5 (which is equivalent to a mean-square maximum of half-normal variables treated by Finch [14]), Kenyon’s isoradial diagonal transfer currents [15], and the mean-field limit of the UST degree law. For the remaining statements our searches found no prior closed form. Novelty should be understood in that limited sense.

II. MODEL SETUP

A. Grover walks on periodic graphs

Let G be a locally finite periodic graph. The walk lives on arcs (directed edges) $a = (o \rightarrow h_a)$, and each vertex v of degree d_v carries the Grover coin $C_v = \frac{2}{d_v}J - I$. In the *flip-flop* walk, the shift sends the arc $(o \rightarrow h)$ to its reverse $(h \rightarrow o)$. In the *moving-shift* walk on $\mathbb{Z}^d$, the coin state $\pm e_i$ is transported by $\pm e_i$ and keeps its label. For

* jacob.l.goodchild@gmail.com

an initial state ψ_0 localized at o , the long-time average return probability is

$$\bar{p} = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^{T-1} \sum_{a: \text{tail}(a)=o} |\langle a | U^t \psi_0 \rangle|^2. \quad (1)$$

After Fourier transformation, the continuous spectrum of $U(k)$ contributes nothing to Eq. (1) by the Riemann–Lebesgue lemma. Only the flat eigenvalues λ (eigenvalues of $U(k)$ that do not depend on k) survive, and

$$\bar{p} = \sum_{\lambda \text{ flat}} \sum_{a \text{ at } o} |\langle a | \bar{\Pi}_\lambda | \psi_0 \rangle|^2, \quad \bar{\Pi}_\lambda = \mathbb{E}_k \Pi_\lambda(k), \quad (2)$$

where $\mathbb{E}_k$ denotes the normalized Brillouin-zone average.

B. Flat bands as cycle spaces: the transfer-current principle

Theorem 1 (transfer-current form of flip-flop trapping). For the flip-flop Grover walk on any periodic graph, starting on arc a at vertex o ,

$$\begin{aligned} \bar{p}(a) = & \frac{1}{4} \sum_{b \text{ at } o} (\delta_{ab} - Y_{ab})^2 + \frac{1}{4} \sum_{b \text{ at } o} (\delta_{ab} - Y_{ab}^Q)^2 \\ & + \sum_{\mu} (\text{flat-band terms} \geq 0). \end{aligned} \quad (3)$$

Here Y_{ab} is the current through arc b when a unit current enters at o and leaves at h_a in a unit-resistor network (the Kirchhoff transfer current). Y^Q is the same object built from $Q = D + A$ in place of $L = D - A$. The last sum runs over flat bands μ of the simple random walk $D^{-1/2}AD^{-1/2}$.

Derivation outline. (1) Let u be the vector with components $d_v^{-1/2}$ on the arcs at v , and let S be the flip-flop shift, so $U = S(2uu^* - I)$. The flat eigenvalue $+1$ is supported on $K = \{\psi : \langle u, \psi \rangle = 0 = \langle u, S\psi \rangle\}$, where $U = -S$. (2) Divergence-free (circulating) flows satisfy both constraints. The $+1$ eigenspace is therefore the cycle space of the graph, and its projector is $\frac{1}{2}(I - S) - \frac{1}{2}(I - S)uu^*(I - S)/\langle u, (I - S)u \rangle$. (3) Averaged over k , the projector onto the orthogonal complement of the cycle space (the cut space) has matrix elements $Y_{ab} = \frac{1}{2}[R(o, h_a) + R(o, h_b) - R(h_a, h_b)]$, written with the resistance distance R , that is, with the Green function $G(x) = \mathbb{E}_k[\cos(k \cdot x)/\sum_r (1 - \cos k \cdot \delta_r)]$. (4) Replacing S by $-S$ interchanges L and Q , which gives the -1 channel. For bipartite graphs Q is unitarily equivalent to L , so the two channels are equal. (5) Every other flat eigenvalue of U comes, through the Szegedy correspondence, from a flat band μ of the random walk, with $\lambda = \mu \pm i\sqrt{1 - \mu^2}$.

Two identities hold throughout:

$$\sum_b Y_{ab} = \sum_b Y_{ab}^Q = 1, \quad (4)$$

$$\sum_{e \in \text{cell}} Y_{ee} = \sum_{e \in \text{cell}} Y_{ee}^Q = |V_{\text{cell}}|. \quad (5)$$

Equation (4) is current conservation. Equation (5) is Foster's theorem [13] per unit cell (the trace of $L^{-1}L$). On an edge-transitive d -regular lattice it forces $Y_{aa} = Y_{aa}^Q = 2/d$.

C. Flip-flop walks on the cubic lattices

For a Bravais lattice with neighbour vectors $\pm\delta_1, \dots, \pm\delta_P$, Eq. (3) becomes $\bar{p}(\phi) = \|M_+\phi\|^2 + \|M_-\phi\|^2$ with $M_+ = \frac{1}{2}I - N$ and

$$N_{(s,p),(t,q)} = \frac{1}{4}[G(0) - G(\delta_p) - G(\delta_q) + G(t\delta_q - s\delta_p)]. \quad (6)$$

Simple cubic. Let

$$W = \frac{\sqrt{6}}{32\pi^3} \Gamma(\frac{1}{24})\Gamma(\frac{5}{24})\Gamma(\frac{7}{24})\Gamma(\frac{11}{24}) = 1.5163860591519780 \dots \quad (7)$$

be Watson's simple-cubic integral [8, 9]. Then $N_{\text{diag}} = 1/6$, $N(+i, -i) = \alpha = (7W + 54/(\pi^2 W) - 12)/36$, and $N(i, j \neq i) = \beta = (24 - 7W - 54/(\pi^2 W))/144$, with $\alpha + 4\beta = 1/3$. Starting in one coin state,

$$\bar{p}_{\text{SC}} = 2\left[\frac{1}{9} + \alpha^2 + 4\beta^2\right] = 0.26672720352085343375 \dots \quad (8)$$

The uniform coin state escapes completely ($\bar{p} = 0$).

Body-centred cubic. With $W_b = \Gamma(\frac{1}{4})^4/(4\pi^3) = 1.39320392968567685918 \dots$, we have $N_{\text{diag}} = 1/8$ and $\alpha = (3W_b + 36/(\pi^2 W_b) - 6)/16$, $\beta_s = (2 - W_b + 4/(\pi^2 W_b))/16$ and $\beta_d = (1 - 8/(\pi^2 W_b))/8$. Hence

$$\bar{p}_{\text{BCC}} = 2\left[\frac{9}{64} + \alpha^2 + 3\beta_s^2 + 3\beta_d^2\right] = 0.32150479124143771009 \dots \quad (9)$$

Face-centred cubic. FCC is not bipartite, so the -1 channel requires the Green function at the opposite band edge, $G'(x) = \mathbb{E}_k[\cos(k \cdot x)/(6 + 2s)]$ with $s = c_1 c_2 + c_2 c_3 + c_3 c_1$ and $c_i = \cos k_i$. We find that the far-edge value is governed by the *simple-cubic* Watson constant:

$$G'(0) = \frac{W}{6\sqrt{2}} = \frac{\sqrt{3}\Gamma(\frac{1}{24})\Gamma(\frac{5}{24})\Gamma(\frac{7}{24})\Gamma(\frac{11}{24})}{192\pi^3}. \quad (10)$$

This identity probably follows from Joyce's analysis of the FCC Green function [11]. With $Y = \sqrt{2}W$ we further have $G'(1, 1, 0) = (2 - Y)/12$, $G'(2, 0, 0) = (7Y + 108/(\pi^2 Y) - 36\sqrt{3}/\pi)/72$, $G'(2, 1, 1) = (2Y - 3 - 27/(\pi^2 Y))/9$, and $G'(2, 2, 0) = (Y + 54/(\pi^2 Y) + 6\sqrt{3}/\pi - 8)/6$. All of these satisfy the FCC lattice equations to a residual of 3×10^{-42} . With

$$W_f = \frac{3\Gamma(\frac{1}{3})^6}{2^{14/3}\pi^4} = 0.44822039439 \dots, \quad (11)$$

the ordinary channel follows from W_f and $1/(\pi^2 W_f)$. The totals are

$$\begin{aligned} \bar{p}_{\text{FCC}} = & 0.18948881307124000416 + 0.18942581798307167926 \\ & = 0.37891463105431168342 \dots \end{aligned} \quad (12)$$

D. Two-dimensional lattices

Triangular. The Laplacian resistances are $a(1) = 1/3$, $a(\sqrt{3}) = 2\sqrt{3}/\pi - 2/3$ and $a(2) = 8/3 - 4\sqrt{3}/\pi$. At the far band edge we find

$$X \equiv \mathbb{E}_k \left[\frac{1}{3 + \cos k_1 + \cos k_2 + \cos(k_1 - k_2)} \right] = \frac{2^{2/3} \Gamma(\frac{1}{3})^3}{8\pi^2}, \quad (13)$$

with $G'(1) = (1 - 3X)/3$, $G'(\sqrt{3}) = (9X - 2 - \sqrt{3}/(\pi X))/3$ and $G'(2) = (3X - 4 + 2\sqrt{3}/(\pi X))/3$. The two channels are

$$(+1) = \frac{43\pi^2 - 120\sqrt{3}\pi + 324}{72\pi^2}, \quad (14)$$

$$(-1) = X^2 - \frac{X}{2} + \frac{19}{72} - \frac{\sqrt{3}}{6\pi} - \frac{\sqrt{3}}{18\pi X} + \frac{1}{8\pi^2 X^2}, \quad (15)$$

with total $\bar{p} = 0.26795570253708055129\dots$. The square lattice gives $\bar{p} = \frac{1}{2} - \frac{\pi}{2} + \frac{3}{\pi^2}$, which is very likely known.

Honeycomb and kagome. On the honeycomb lattice, Eq. (5) and edge transitivity force $Y = (2/3, 1/6, 1/6)$, so $\bar{p} = 1/12$ exactly. Kagome has a flat band $\mu = -1/2$ of the random walk. This produces two extra walk eigenvalues $\lambda = e^{\pm 2\pi i/3}$, each contributing $(19 - 42\sqrt{3}/\pi + 108/\pi^2)/648$. With $K = K(m)$ and $E = E(m)$ at parameter $m = 5/32$ (we use the parameter convention $K(m) = \int_0^{\pi/2} (1 - m \sin^2 \theta)^{-1/2} d\theta$ throughout), the total is

$$\begin{aligned} \bar{p}_{\text{kag}} &= \frac{103}{144} - \frac{7\sqrt{3}}{48\pi} + \frac{3}{8\pi^2} - \frac{23\sqrt{2}E}{12\pi} + \frac{127\sqrt{2}K}{288\pi} \\ &\quad + \frac{4E^2}{\pi^2} - \frac{13KE}{6\pi^2} + \frac{61K^2}{144\pi^2} \\ &= 0.19146243000085471390\dots \end{aligned} \quad (16)$$

The elliptic input is the identity $\mathbb{E}_k[1/(11 - \cos k_1 - \cos k_2 - \cos(k_1 - k_2))] = K(5/32)/(4\sqrt{2}\pi)$.

Lieb, dice, 4.8.8, checkerboard and star. Table I collects these results. We write $c = \sqrt{2} \arccos(1/3)/\pi$. The 4.8.8 Laplacian determinant factorizes as $64 - 4(3 + \cos k_1)(3 + \cos k_2)$, which is the origin of $\arccos(1/3)$. For the star lattice, $\kappa = \sqrt{6} K(3/128)/\pi$.

E. Line graphs

Let $L(G)$ be the line graph of a Bravais lattice G with bond vectors $e_1, \dots, e_P$, and put $B(k) = (1 + e^{ik \cdot e_t})_{t=1}^P$. Then $A_{L(G)} + 2 = B^\dagger B$, the degree is $4P - 2$, and

$$L^{-1} = \frac{1}{4P} [I + B^\dagger B / (4P - BB^\dagger)], \quad (17)$$

$$Q^{-1} = \frac{1}{4P-4} [I - B^\dagger B / (4P - 4 + BB^\dagger)], \quad (18)$$

$$F = I - B^\dagger B / (BB^\dagger), \quad (19)$$

where F is the projector onto the flat band $\mu = -2/(4P - 2)$. Every trapping channel on $L(G)$ is therefore a fi-

nite combination of Green functions of G at three energies. The checkerboard ($= L(\mathbb{Z}^2)$) and kagome ($= L(\text{honeycomb})$) are instances.

Line graph of $\mathbb{Z}^3$ (degree 10). Let $g(t) = \mathbb{E}[1/(t - \cos k_1 - \cos k_2 - \cos k_3)]$ and $g_j = g^{(j)}(7)$. Every channel is a polynomial in W , $X = 1/(\pi^2 W)$ and the g_j . For the collinear start,

$$\begin{aligned} (+1) &= \frac{65}{324} - \frac{109W}{2592} - \frac{9X}{16} + \frac{45701W^2}{2985984} + \frac{9347WX}{27648} + \frac{1973X^2}{1024}, \\ (\text{flat}) &= \frac{259}{1296} - \frac{6233W}{41472} - \frac{453X}{256} + \frac{773765W^2}{23887872} \\ &\quad + \frac{166595WX}{221184} + \frac{37685X^2}{8192}, \end{aligned} \quad (20)$$

and $\bar{p} = 0.38013642908365187096$ (perpendicular start: 0.38100581744260323594). Joyce's closed form for g [10] makes the -1 channel a polynomial in complete elliptic integrals.

Line graphs of BCC and FCC and of the triangular lattice. These are solved completely in the same way. The totals are listed in Table II. The BCC case needs, besides W_b , the Green function $\mathbb{E}[1/(10 + 4c_1 c_2 c_3)] = \kappa^2/10$ with $\kappa = 2K(m)/\pi$ and $m = \frac{1}{2} - \sqrt{21}/10$. The triangular case needs the Green function *below* the band. Its modulus is complex, but a modular transformation gives the real form

$$\mathbb{E}_k \left[\frac{1}{7 + \cos k_1 + \cos k_2 + \cos(k_1 - k_2)} \right] = \frac{K(m_*)}{135^{1/4}\pi}, \quad (21)$$

where $m_* = \frac{1}{2} - \frac{23\sqrt{15}}{180}$ is a root of $2160m^2 - 2160m + 11$.

F. Pyrochlore

The pyrochlore lattice ($= L(\text{diamond})$, degree 6) has a doubly degenerate flat band $\mu = -1/3$. Its projector is $F = I - B^\dagger (BB^\dagger)^{-1} B$, where B is the diamond incidence matrix. With $W = W_f$ and $X = 1/(\pi^2 W)$, the Laplacian transfer currents from an edge are $1/3$ (self), $1/6$ (the two other edges of the same tetrahedron), $(1 + 8W + 3X)/48$ (collinear partner) and $(15 - 8W - 3X)/96$ (the other two). The channels are

$$(+1) = \frac{W^2}{96} + \frac{WX}{128} - \frac{13W}{1152} + \frac{3X^2}{2048} - \frac{13X}{3072} + \frac{2531}{18432}, \quad (22)$$

$$(\text{flat}) = \frac{3W^2}{64} + \frac{9WX}{256} - \frac{13W}{256} + \frac{27X^2}{4096} - \frac{39X}{2048} + \frac{163}{4096}. \quad (23)$$

The -1 channel is a quadratic polynomial in g_0, g_1, g_2 , the FCC Green function $g(t) = \mathbb{E}[1/(t - s)]$ and its first two derivatives at $t = 15$, which Joyce's formula makes explicit. The total is $\bar{p}_{\text{pyro}} = 0.29388359496\dots$

G. Anisotropic walks

Square lattice. Take the random walk with $P(\pm x) = \alpha/2$ and $P(\pm y) = \beta/2$ ($\alpha + \beta = 1$), its Szegedy walk, and $\theta = \arcsin \sqrt{\alpha}$. The key integral is $\mathbb{E}[(1 - \cos k_1)/(\alpha(1 -$

TABLE I. Exact long-time return probabilities of the flip-flop Grover walk on two-dimensional lattices, from Eq. (3). $c = \sqrt{2} \arccos(1/3)/\pi$, $\kappa = \sqrt{6}K(3/128)/\pi$, and K, E in the checkerboard rows are at parameter $m = 1/4$.

Lattice, start	Closed form	Value
Lieb, corner	$(1 - 4/\pi + 6/\pi^2)/4$	0.0836718892797159856
Lieb, edge site	$1/8$	0.125
Dice, hub	$(31 - 20\sqrt{3}/\pi + 45/\pi^2)/144$	0.17036718996853721
Dice, rim	$11/48$	0.22916
4.8.8, square edge	$(4 - 6c + 3c^2)/16$	0.0997756085862333247
4.8.8, bridge edge	$3c^2/16$	0.0575730175785728199
Star 3.12.12, triangle edge	$1883/10800 - 7\kappa/96 + 3\kappa^2/128$	0.1200922563314837021
Star 3.12.12, bridge edge	$107/2700 + \kappa/48 + 3\kappa^2/128$	0.1008716548141747862
Checkerboard, collinear	$\frac{2097}{1024} - \frac{185}{128\pi} + \frac{79}{32\pi^2} + \frac{123K}{32\pi} - \frac{93E}{8\pi} + \frac{39K^2}{16\pi^2} - \frac{57KE}{4\pi^2} + \frac{43E^2}{2\pi^2}$	0.2915287743505391057
Checkerboard, perpendicular	$\frac{777}{1024} - \frac{37}{64\pi} + \frac{25}{32\pi^2} + \frac{45K}{64\pi} - \frac{33E}{16\pi} + \frac{3K^2}{4\pi^2} - \frac{3KE}{\pi^2} + \frac{7E^2}{2\pi^2}$	0.2954779129769429679

TABLE II. Flip-flop trapping on line graphs (start classes by the angle between the start bond and the site's own bond).

Line graph	start	$\bar{p}$
$L(\text{triangular})$	collinear	0.37891749229289535007
	60°	0.37983613893989158023
	120°	0.38314651208017409854
$L(\text{SC})$	collinear	0.38013642908365187096
	perpendicular	0.38100581744260323594
$L(\text{BCC})$	collinear	0.41704934756699764035
	70.5°	0.41732104849596726107
	109.5°	0.41775946732413853940
$L(\text{FCC})$	collinear	0.44924601047168703813
	60°	0.44932628436614800807
	90°	0.44943990277365356359
	120°	0.44977663121249284644

$\cos k_1) + \beta(1 - \cos k_2)] = 2\theta/(\pi\alpha)$. It gives

$$Y(+x, +x) = 2\theta/\pi, \quad Y(+x, -x) = \frac{2}{\pi}(\cot \theta - \theta \cot^2 \theta), \quad (24)$$

$$Y(+x, \pm y) = \frac{\tan \theta}{2} \left[1 - \frac{2}{\pi}(\theta + \cot \theta - \theta \cot^2 \theta) \right], \quad (25)$$

and $\bar{p}(\alpha) = \frac{1}{2}[(1 - 2\theta/\pi)^2 + Y(+x, -x)^2 + 2Y(+x, y)^2]$. As checks, $\alpha = 1/4$ gives 0.25400036756659953284 and $\alpha = 0.7$ gives 0.10771461463761465554.

Honeycomb. Choose bond type i with probability p_i , and set $\theta_i = \arctan(\lambda p_i)$ with $\lambda = (p_1 p_2 p_3)^{-1/2}$, so that $\sum \theta_i = \pi$. Write $t_i = \theta_i/\pi$. Then $T(1, 1) = 2t_1$ (Kenyon [15]) and

$$T(1, j) = \left[\frac{1}{2} - p_k - p_1 t_1 - p_j t_j + p_k t_k \right] / p_1, \quad \{j, k\} = \{2, 3\}, \quad (26)$$

so that $\bar{p} = \frac{1}{2}[(1 - 2t_1)^2 + (p_1/p_2)T(1, 2)^2 + (p_1/p_3)T(1, 3)^2]$. For example, $p = (1/2, 1/4, 1/4)$ gives $\bar{p} = 0.04680597938234766442$.

Triangular. For conductances q_i with $\sum q_i = 1$, set $\theta_i = \arctan(\lambda q_i)$ with $\lambda = (q_1 q_2 + q_2 q_3 + q_3 q_1)^{-1/2}$, so

that $\sum \theta_i = \pi/2$. Let $t_i = \theta_i/\pi$ and $c_1 = \cot \theta_1$. The six currents at the origin for a unit current from o to its $+e_1$ neighbour are

$$\begin{aligned} T(+e_1) &= 2t_1, & T(-e_1) &= \frac{2}{\pi}(c_1 - \theta_1 c_1^2), \\ T(+e_2) &= -\frac{q_2}{2q_3} + \frac{q_2(q_1+q_3)}{q_1 q_3} t_1 + \frac{q_2+q_3}{q_3} t_2, \\ T(-e_2) &= \frac{q_2(q_1+q_3)}{q_1^2} t_1 + \frac{q_2+q_3}{q_2} t_2 - \frac{c_1}{\pi}, \\ T(+e_3) &= \frac{1}{2} - \frac{q_1-q_3}{q_1} t_1 - \frac{q_2+q_3}{q_2} t_2, \end{aligned} \quad (27)$$

and $T(-e_3) = 1 - \sum(\text{others})$. The diagonal is Kenyon's isoradial theorem [15].

H. Lazy (Szegedy) flip-flop walks

Add a self-loop with holding probability ε and use the Szegedy coin. Self-loops carry no circulation, so the $+1$ channel is unchanged, while the -1 channel acquires a loop amplitude.

Square lattice. Let $k = (1 - \varepsilon)/(1 + \varepsilon)$, $\mathcal{K} = K(k^2)/\pi$ and $\mathcal{E} = E(k^2)/\pi$. The amplitudes are $A(+x) = \frac{1}{4} + \frac{1}{2}(1 - k)\mathcal{K}$, $A(-x) = -\frac{1}{4}[1 - 2/k + 2(k - 1)\mathcal{K} + 4\mathcal{E}/k]$, $A(\pm y) = -\frac{1}{4}[1 + (2/k - 2)\mathcal{K} - 2\mathcal{E}/k]$ and $A(\text{loop}) = -\sqrt{(1 - k)/(2k)}(\frac{1}{2} - (1 - k)\mathcal{K})$. Then

$$\bar{p}(\varepsilon) = \frac{1}{4} - \frac{1}{\pi} + \frac{3}{2\pi^2} + A(+x)^2 + A(-x)^2 + 2A(y)^2 + A(\text{loop})^2. \quad (28)$$

Triangular lattice. Let $t = 3(1 + \varepsilon)/(1 - \varepsilon)$ and $S = \cos k_1 + \cos k_2 + \cos(k_1 - k_2)$. For $t > 3/2$ the Green function has the real-parameter form

$$g(t) = \mathbb{E}_k \left[\frac{1}{t + S} \right] = \frac{2K(m)}{\pi[(t - 1)^3(t + 3)]^{1/4}}, \quad (29)$$

$$m = \frac{1}{2} - \frac{t^2 - 3}{2(t - 1)^{3/2}\sqrt{t + 3}}. \quad (30)$$

We need $G_1 = (1 - tg)/3$, together with

$$G_{\sqrt{3}} = -\frac{2}{3} + \frac{2t^2+3t-6}{3}g + \frac{(t-1)(t+3)(2t-3)}{3}g', \quad (31)$$

$$G_2 = -\frac{2(t-1)}{3} - \frac{2t^2+4t-9}{3}g - \frac{2(t-1)(t+3)(2t-3)}{3}g'. \quad (32)$$

The -1 amplitudes are $A(+e_1) = \frac{1}{3} + (t-3)g/6$, $A(-e_1) = -(g+2G_1+G_2)/4$, $A(\text{adj}) = -(g+3G_1)/4$ (two arcs), $A(\text{rem}) = -(g+2G_1+G_{\sqrt{3}})/4$ (two arcs) and $A(\text{loop}) = -\frac{\sqrt{t-3}}{6}(1+(3-t)g)$, and

$$\begin{aligned} \bar{p}(\varepsilon) = & \frac{43\pi^2 - 120\sqrt{3}\pi + 324}{72\pi^2} + A(+e_1)^2 + A(-e_1)^2 \\ & + 2A(\text{adj})^2 + 2A(\text{rem})^2 + A(\text{loop})^2. \end{aligned} \quad (33)$$

For example, $\bar{p}(0.2) = 0.31653136185695229253$ and $\bar{p}(0.5) = 0.35388644533661557374$.

Honeycomb. Let $r = 3(1+\varepsilon)/(1-\varepsilon)$, $m = 16r/[(r-1)^3(r+3)]$ and $g = 4K(m)/(\pi\sqrt{(r-1)^3(r+3)})$. Then

$$\begin{aligned} 288\bar{p}(\varepsilon) = & g^2r^2(r-3)^2(r-1)(r+3) \\ & - 4gr(r-4)(r-3)(r+3) + 4(r^2-4r+9). \end{aligned} \quad (34)$$

At $\varepsilon = 0$ this gives $1/12$.

Diamond. Let $r = 4(1+\varepsilon)/(1-\varepsilon)$ and $\tau = (r^2-4)/4$, and let $g = \mathbb{E}[1/(\tau-s)]$ be the FCC Green function in Joyce's closed form [11], $g = \tau^{-1}\mathcal{L}(3/\tau)$ with $\mathcal{L}(x) = 3^{3/2}(3+x)^{-3/2}(2-\sqrt{1-x})(2K(m)/\pi)^2$ and $m = \frac{1}{2} - \frac{\sqrt{3}}{2}(3+x)^{-3/2}(4x+(3-x)\sqrt{1-x})$. Then

$$\begin{aligned} 3072\bar{p}(\varepsilon) = & g^2r^2(r-4)^2(r-2)(r+4) \\ & - 8gr(r-6)(r-4)(r+4) + 16(r^2-6r+40). \end{aligned} \quad (35)$$

At $\varepsilon = 0$ this gives $1/6$.

General z . For a z -regular bipartite lattice of girth at least 6 whose vertex symmetry acts 2-transitively on the arcs, let $a_2(k) = |f(k)|^2 - z$, $r = z(1+\varepsilon)/(1-\varepsilon)$, $h = \mathbb{E}[1/(r^2-z-a_2)]$ and $u = hr(r-z)$. The amplitudes are $A_{\text{start}} = (2u+z-2)/(2z)$, $A_{\text{other}} = -[u(r-z+2) - r+2z-2]/[2z(z-1)]$ and $A_{\text{loop}}^2 = (r-z)(1-u)^2/(2z^2)$, and

$$\bar{p}(\varepsilon) = \frac{(z-2)^2}{4z(z-1)} + A_{\text{start}}^2 + (z-1)A_{\text{other}}^2 + A_{\text{loop}}^2. \quad (36)$$

III. THE UNIVERSAL LOCALIZATION BOUND THEOREM

Theorem 2. Consider the flip-flop Grover walk on a periodic graph, started on an arc a at a vertex of degree d . Then

$$\bar{p}(a) \geq \frac{d}{4(d-1)} \left[(1 - Y_{aa})^2 + (1 - Y_{aa}^Q)^2 \right]. \quad (37)$$

If the lattice is edge-transitive and d -regular, where d is the *vertex degree* and not the spatial dimension, then

$$\bar{p} \geq \frac{(d-2)^2}{2d(d-1)}. \quad (38)$$

Proof. The flat-band terms in Eq. (3) are nonnegative, so it suffices to bound each ± 1 channel. By Eq. (4), the $d-1$ off-diagonal entries Y_{ab} ($b \neq a$) sum to $1 - Y_{aa}$. The Cauchy-Schwarz inequality then gives $\sum_{b \neq a} Y_{ab}^2 \geq (1 - Y_{aa})^2/(d-1)$. Hence

$$\begin{aligned} \frac{1}{4} \sum_b (\delta_{ab} - Y_{ab})^2 & \geq \frac{1}{4} \left[(1 - Y_{aa})^2 + \frac{(1 - Y_{aa})^2}{d-1} \right] \\ & = \frac{d}{4(d-1)} (1 - Y_{aa})^2, \end{aligned} \quad (39)$$

and likewise for Y^Q . Under edge transitivity, Eq. (5) gives $Y_{aa} = Y_{aa}^Q = 2/d$, and substituting yields Eq. (38). $\square$

Equality. Equality holds if and only if (i) all off-diagonal currents from a are equal in both channels and (ii) there are no additional flat bands. Both hold for bipartite lattices whose vertex stabilizer permutes the d arcs 2-transitively, because then Y^Q equals Y up to signs. This gives the exact values

$$\begin{aligned} \text{honeycomb } (d=3) : & \quad \bar{p} = 1/12, \\ \text{Laves graph / srs } (d=3) : & \quad \bar{p} = 1/12, \\ \text{diamond } (d=4) : & \quad \bar{p} = 1/6, \\ d\text{-dim. diamond } (D=d+1) : & \quad \bar{p} = \frac{(d-1)^2}{2d(d+1)}. \end{aligned} \quad (40)$$

In particular, 4D diamond has $\bar{p} = 9/40$. Brillouin-zone averages of the walk's own eigenprojectors give 0.16650 (12^3 grid) and 0.16660 (16^3) for diamond, and 0.22490 (8^4) for 4D diamond.

Tightness elsewhere. Comparing with the exact values of Sec. II: square 0.1673438 vs. $1/6$; triangular 0.2679557 vs. $4/15$; simple cubic 0.2667272 vs. $4/15$; BCC 0.3215048 vs. $9/28$; FCC 0.3789146 vs. $25/66$. The bound is within 10^{-3} in every case we examined, and within 10^{-4} for simple cubic and BCC.

IV. HIGHER-DIMENSIONAL TRAPPING INTEGRALS

A. Moving-shift Grover walk on $\mathbb{Z}^d$

Consider the Grover coin $G = \frac{1}{d}J - I$ on the $2d$ coin states $\pm e_i$ with the moving shift. In momentum space $S(k)G$ has eigenvalues ± 1 for every k . Solving $G\psi = \lambda S(k)^*\psi$ gives flat-band components $(1 \pm i \tan \frac{k_i}{2})/2$ for $\lambda = +1$ and $(1 \mp i \cot \frac{k_i}{2})/2$ for $\lambda = -1$. When k_i is uniform on the circle, $\tan(k_i/2)$ and $\cot(k_i/2)$ are standard Cauchy variables t_i . Hence

$$\overline{M} = \frac{A}{2}J + \frac{B}{2} \text{blockdiag} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad (41)$$

with

$$A_d = \mathbb{E}\left[\frac{1}{d + \sum_i t_i^2}\right] = \int_0^\infty \operatorname{erfc}(\sqrt{s})^d ds, \quad B_d = \frac{1 - dA_d}{d}. \quad (42)$$

The second form of A_d follows from $1/x = \int_0^\infty e^{-sx} ds$ and $\mathbb{E}[e^{-st^2}] = e^s \operatorname{erfc} \sqrt{s}$. With $u = (1, \dots, 1)/\sqrt{2d}$ and P_{anti} the projector onto $\{e_{+i} - e_{-i}\}$,

$$\bar{p}(\phi) = 2[(dA_d)^2 \langle u|\phi \rangle^2 + B_d^2 \|P_{\text{anti}}\phi\|^2], \quad (43)$$

so that $\bar{p}(+e_1) = dA_d^2 + B_d^2$ and $\bar{p}(u) = 2d^2 A_d^2$. The exact values are

$$A_2 = \frac{1}{2} - \frac{1}{\pi}, \quad (44)$$

$$A_3 = \frac{1}{2} - \frac{3-\sqrt{3}}{\pi} = 0.09639923687042003490\dots, \quad (45)$$

$$A_4 = \frac{1}{2} - \frac{18-8\sqrt{3}}{3\pi} = 0.06035107068870143614\dots, \quad (46)$$

$$A_5 = \frac{1}{2} - \frac{10}{\pi} + \frac{40\sqrt{3}}{3\pi} - \frac{20\sqrt{3} \arccos(1/4)}{\pi^2} = 0.04153865644854222924\dots \quad (47)$$

The $d = 2$ case reproduces Inui *et al.* [4]. The 3D headline values are $\bar{p}(+e_1) = 0.08401620467429515245$ and $\bar{p}(u) = 0.1672706316455882907$; in 5D, $\bar{p}(+e_1) = 0.03373729729788322185$. Because $A_d = \frac{1}{2}\mathbb{E}[\min_k Z_k^2]$ for independent standard normals, min/max inclusion-exclusion turns A_5 into $\mathbb{E}[\max_{k \leq 5} Z_k^2] = 1 + 20\sqrt{3}/\pi - 40\sqrt{3} \arccos(1/4)/\pi^2$. That is a mean-square maximum of half-normals, the setting considered by Finch [14], so the integral itself is probably known. A_5 closes because it is a moment over a region of the even-dimensional sphere S^4 . For $d = 6$ (S^5) no closed form was found.

B. Weighted coins: a general- d theorem

Replace the Grover coin by $2|s\rangle\langle s| - I$ with $|s\rangle = (\sqrt{p_1/2}, \sqrt{p_1/2}, \dots, \sqrt{p_d/2}, \sqrt{p_d/2})$ and $\sum p_i = 1$. Let Z_i be independent standard normals and define

$$A = \frac{1}{2}\mathbb{E}\left[\min_k Z_k^2/p_k\right] = \int_0^\infty \prod_k \operatorname{erfc}(\sqrt{p_k}u) du, \quad (48)$$

together with $\pi_i = P(\text{coordinate } i \text{ attains the minimum})$.

Theorem 3. Starting in coin state $+e_1$,

$$\bar{p} = 2\left[\frac{\pi_1^2}{4} + \left(p_1 A - \frac{\pi_1}{2}\right)^2 + \frac{p_1(1-p_1)A^2}{2}\right]. \quad (49)$$

The proof uses the Cauchy identities $\mathbb{E}[1/(1 + \sum pt^2)] = A$ and $\mathbb{E}[t_i^2/(1 + \sum pt^2)] = \pi_i/p_i - A$.

In $d = 3$ everything is elementary. For $\{i, j, k\} = \{1, 2, 3\}$,

$$\pi_i = \frac{2}{\pi} \left[\arctan \sqrt{\frac{p_i}{p_j}} + \arctan \sqrt{\frac{p_i}{p_k}} - \arctan \sqrt{\frac{p_i}{p_j p_k}} \right], \quad (50)$$

$$A = \frac{1}{2} \sum_i \left[\frac{\pi_i}{p_i} - \frac{2}{\pi} \left(\sqrt{\frac{p_k/p_i}{1+\sqrt{p_j}}} + \sqrt{\frac{p_j/p_i}{1+\sqrt{p_k}}} \right) \right]. \quad (51)$$

These are obtained from solid angles of the three cones meeting at $(\sqrt{p_1}, \sqrt{p_2}, \sqrt{p_3})$ together with Stokes' theorem for $\int_T x_i^2 d\sigma$. For example, $p = (0.2, 0.3, 0.5)$ gives $\bar{p} = 0.05362025915918515966$. The $d = 2$ case may overlap earlier work on generalized Grover coins.

C. Lackadaisical moving-shift walk

Add a self-loop of weight l to the moving-shift walk, so that $|s\rangle = (1, \dots, 1, \sqrt{l})/\sqrt{2d+l}$ [3]. The -1 flat band disappears. With

$$a(l) = \frac{1}{2} \int_0^\infty e^{-ls/2} \operatorname{erfc}(\sqrt{s})^d ds, \quad b = \frac{1 - (2d+l)a}{2d}, \quad (52)$$

the return probability is $l(2d+l)a^2$ from the loop state and $2da^2 + 2b^2 + la^2$ from a moving state. The closed forms are

$$d = 2: \quad a = \frac{1}{l} \left[1 - \frac{4}{\pi} \frac{\arctan q}{q} \right], \quad q = \sqrt{1+l/2}, \quad (53)$$

$$d = 3: \quad a = \frac{1}{l} \left[1 - 3\sqrt{\frac{2}{l+2}} \left(1 + \frac{2}{\pi} \left(\arcsin \frac{2}{l+4} - 2 \arcsin \sqrt{\frac{2}{l+4}} \right) \right) \right]. \quad (54)$$

The $d = 3$ case rests on the lemma

$$\int_0^\infty e^{-bx^2} \operatorname{erfc}(x)^2 dx = \frac{1}{2} \sqrt{\frac{\pi}{b}} \left[1 + \frac{2}{\pi} \left(\arcsin \frac{1}{b+1} - 2 \arcsin \frac{1}{\sqrt{b+1}} \right) \right], \quad (55)$$

which is proved via a trivariate Gaussian orthant probability.

D. Moving-shift triangular walk (partial)

On the triangular lattice with the moving shift, the -1 channel from $+e_1$ is $\frac{5}{12} - \frac{4\sqrt{3}}{3\pi} + \frac{57}{16\pi^2} = 0.04251818949677224475$. The $+1$ channel is 0.04312949681537029391 , for which no closed form was found in bases built from π , $\sqrt{3}$, logarithms, Clausen values, Catalan's constant and $\arccos(1/3)$. The total is $\bar{p} = 0.08564768631214253866$.

V. UST DEGREE LAWS

Let Y be the $z \times z$ transfer-current matrix of the edges at a vertex. By the transfer-current theorem of Burton and Pemantle [7] and multilinearity of the determinant, the degree of the vertex in the uniform spanning tree satisfies

$$\sum_k P(\deg = k) x^k = \det[I + (x-1)Y]. \quad (56)$$

The method reproduces the square-lattice values of Manna, Dhar and Majumdar [16].

Theorem 4 (binomial law). Suppose the lattice is edge-transitive and the vertex symmetry acts 2-transitively on its z edges (honeycomb, diamond, Laves graph, d -dimensional diamond). Then

$$\deg \stackrel{d}{=} 1 + \text{Binomial}\left(z-1, \frac{1}{z-1}\right). \quad (57)$$

Proof. 2-transitivity forces $Y = aI + b(J - I)$. Foster's theorem gives $a = 2/z$, and Eq. (4) gives $a + (z-1)b = 1$. So Y has eigenvalue 1 on the all-ones vector and $a - b = 1/(z-1)$ with multiplicity $z-1$. Hence $\det[I + (x-1)Y] = x\left(\frac{z-2}{z-1} + \frac{x}{z-1}\right)^{z-1}$. $\square$

For the honeycomb this gives $(1/4, 1/2, 1/4)$ and for diamond $(8/27, 4/9, 2/9, 1/27)$. As $z \rightarrow \infty$ the law tends to $1 + \text{Poisson}(1)$, the known mean-field limit. *The law does not hold* for the pyrochlore, king or snub-square lattices, whose vertex symmetry is not 2-transitive on the edges. Their distributions are given below.

Simple cubic. With $\alpha = (7W + 54/(\pi^2 W))/36 = 0.39507915234185137202$,

$$\begin{aligned} P(1) &= 8\alpha^3(3\alpha - 2)^2, \\ P(2) &= -4\alpha^2(3\alpha - 2)(30\alpha^2 - 25\alpha + 6), \\ P(3) &= 2\alpha(360\alpha^4 - 600\alpha^3 + 379\alpha^2 - 108\alpha + 12), \\ P(4) &= -(2\alpha - 1)(360\alpha^4 - 480\alpha^3 + 241\alpha^2 - 52\alpha + 4), \\ P(5) &= 2(2\alpha - 1)^2(3\alpha - 1)(15\alpha^2 - 10\alpha + 2), \\ P(6) &= -(2\alpha - 1)^3(3\alpha - 1)^2. \end{aligned} \quad (58)$$

Numerically these are 0.327494821326, 0.409830450502, 0.204859445598, 0.0511275258041, 0.00637070295715 and 0.000317053812167.

BCC and FCC. With $q = 1/(\pi^2 W)$ for the respective Watson constant,

$$\begin{aligned} P_{\text{BCC}}(1) &= \frac{3q}{16}(W + 4q)^3(4 - W - 12q)^3 \\ &= 0.339633990106590943, \\ P_{\text{BCC}}(8) &= \frac{1}{64}(12q - 1)(W + 4q - 2)^3(W + 12q - 2)^3 \\ &= 1.17884249933455210 \times 10^{-6}, \\ P_{\text{FCC}}(1) &= \frac{1}{5038848}(5 - 4W - 6q)^2(7 - 8W - 3q)^3 \\ &\quad \times (8W + 3q + 1)^3(16W + 15q - 5)^3 \\ &= 0.349518246140286417. \end{aligned} \quad (60)$$

The intermediate degrees are polynomials in W and q .

Pyrochlore ($W = W_f$, $X = 1/(\pi^2 W)$):

$$\begin{aligned} P(1) &= \frac{1}{14155776}[(31 - 8W - 3X)(67 + 24W + 9X)]^2 \\ &= 0.321497700833733694, \\ P(6) &= \frac{1}{28311552}[(8W + 3X + 1)(24W + 9X - 29)]^2 \\ &= 0.000257114517586699387. \end{aligned} \quad (61)$$

King lattice ($p = 1/\pi$, $L = \ln(2 + \sqrt{3})/(\pi\sqrt{3})$):

$$\begin{aligned} P(8) &= \frac{1}{128}(1 + 2p - 7L)(4p - 1 - 2L)(6L + 12p - 5) \\ &\quad \times (1 + 20L - 12L^2 - 32p + 48p^2)^2 \\ &= 8.81169962728673452 \times 10^{-7}. \end{aligned} \quad (62)$$

Snub square ($p = 1/\pi$), all degrees:

$$\begin{aligned} P(1) &= \frac{5}{3888}[8953 - 5058\sqrt{3} + (6336\sqrt{3} - 10692)p \\ &\quad + (1836 - 1296\sqrt{3})p^2], \\ P(2) &= \frac{1}{972}[30245\sqrt{3} - 51893 + (52758 - 30708\sqrt{3})p \\ &\quad + (5940\sqrt{3} - 9666)p^2], \\ P(3) &= \frac{1}{648}[56447 - 32500\sqrt{3} + (28872\sqrt{3} - 50112)p \\ &\quad + (9504 - 5400\sqrt{3})p^2], \\ P(4) &= \frac{1}{972}[32055\sqrt{3} - 55511 + (45522 - 26172\sqrt{3})p \\ &\quad + (4860\sqrt{3} - 8694)p^2], \\ P(5) &= \frac{1}{3888}[50057 - 28910\sqrt{3} + (22608\sqrt{3} - 38988)p \\ &\quad + (7236 - 4320\sqrt{3})p^2]. \end{aligned} \quad (63)$$

Further two-dimensional laws. Again $c = \sqrt{2} \arccos(1/3)/\pi$ and $r = \sqrt{3}/\pi$. The 4.8.8 lattice has $P(1) = P(3) = 3c(4 - 3c)/16$ and $P(2) = (9c^2 - 12c + 8)/8$. The star lattice has $P = \frac{(5+3r)(17-6r)}{324}, \frac{18r^2-21r+158}{324}, \frac{18r^2-21r+77}{324}, \frac{(4-3r)(1+6r)}{324}$, with $P(1) - P(4) = P(2) - P(3) = \frac{1}{4}$ exactly. The dice rim site and the triakis triangle centre are themselves binomial: $1 + \text{Bin}(2, \frac{1}{4})$ and $1 + \text{Bin}(2, \frac{1}{10})$. For the triangular lattice, $P(1) = (3r - 1)(6r - 5)^2(6r + 1)^2/108$, consistent with the known triangular sandpile height-one probability [17]. The general determinantal approach of Rapoport [18] covers such degree statistics in principle. In every case the distribution sums to 1 and has mean exactly 2, and $P(1)$ equals z times the sandpile height-one probability (the companion paper).

VI. NUMERICAL AND SYMBOLIC VERIFICATION METHODOLOGY

Exact Bloch engine. For a two-dimensional periodic graph with Bloch Laplacian $L(k)$, we evaluate every needed average $\mathbb{E}_k[\cdot]$ by an exact residue sum in $z_2 = e^{ik_2}$ (the roots of $\det L$, polished by Newton iteration, with a contour integral around clustered roots near the node at $k = 0$), followed by $\tanh$ - $\sinh$ quadrature in k_1 at 25–60 digits. The three-dimensional engine integrates k_3 exactly and the remaining two dimensions by quadrature. It was validated on the simple-cubic Watson integral to 12 digits, and BCC Green functions were computed by accelerated exact binomial series to 45 digits.

Integer-relation identification. Closed forms were found with PSLQ [19] (mpmath implementation) over

small bases. Relations were accepted only when the coefficients were small and the relation was stable under an increase of precision. Representative precisions: A_3, A_4 at 60 digits; A_5 at 80 digits (residual $< 10^{-80}$); simple-cubic currents at 40 digits (rechecked at 70); FCC at 32 and 50 digits; kagome and triangular at 35–45 digits; the line graph of FCC at 34 digits, with the energy-8 relations re-verified at 60 digits. In that last case one 34-digit relation proved *spurious* and was corrected. This cautions against accepting PSLQ output at marginal precision.

Independent checks. (a) Symbolic assembly of each channel was compared with the engine’s direct value (typically 20–40 digits). (b) Brillouin-zone diagonalization of the full walk operator $U(k)$ on grids (64^3 for the moving shift, 36^3 for $L(\text{SC})$, 90×90 for kagome), averaging each flat projector. (c) Direct simulation of the walk. Examples: square 0.16734263 vs. 0.16734378; triangular 0.2679596 vs. 0.2679557; honeycomb 0.083337 vs. $1/12$; lazy triangular $\varepsilon = 0.2$ 0.316536 vs. 0.316531. 3D simulations are memory-limited to $t \lesssim 90$, and their residual decays like $t^{-3/2}$, as expected for the escaping part of the wave. (d) Closed-form Green functions were checked against the lattice equations, for example $6G'(x) + \frac{1}{2} \sum_{\text{nbrs}} G'(x+e) = \delta_{x,0}$ on FCC with residual 3×10^{-42} . (e) Every UST distribution was checked to sum to 1 with mean exactly 2 (to 10^{-16}), and against the independent sandpile determinant.

VII. DISCUSSION AND CONCLUSION

The transfer-current principle, Eq. (3), turns Grover-walk localization into a problem of classical electrical net-

works. This explains why the trapping probabilities involve exactly the constants of lattice potential theory: Watson integrals, Joyce’s elliptic forms, and Horiguchi’s triangular Green function. It also explains why the signless Laplacian, that is, the Green function at the *opposite* band edge, governs the -1 channel of non-bipartite lattices. A by-product is the far-edge FCC identity, Eq. (10), and the real-parameter triangular Green function, Eq. (30).

The universal bound, Theorem 2, is sharp on 2-arc-transitive bipartite lattices and within 10^{-3} of every exact value we computed. So localization in flip-flop Grover walks is controlled, to excellent accuracy, by the vertex degree alone. The same symmetry yields the binomial UST degree law of Theorem 4, which connects quantum-walk localization to spanning-tree combinatorics.

Open problems include a closed form for the $+1$ channel of the moving-shift triangular walk, the integral A_6 , and the -1 channel of the anisotropic triangular walk. The latter needs a signless Green function whose elliptic parameter depends on the weights.

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